



Decision Support

Profiting from product-harm crises in competitive markets

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ARTICLE INFO

Article history:

Received 9 September 2016

Accepted 7 July 2017

Available online 25 July 2017

Keywords:

Marketing

Pricing

Product recall

Stochastic optimal control

Stochastic differential games

ABSTRACT

Product-harm crises force managers to implement pricing strategies that optimally balance profitability with future potential costs of a recall. We propose a stochastic differential game to investigate what should be the optimal competitive pricing strategies in such contexts. In the proposed model, firms strategically choose prices taking into account that any player in the market can be forced to recall large volume of products at random dates. The characterization of the optimal pricing strategies yields novel managerial insights. We first find that prices of risk-prone products should be higher than prices of risk-free products due to the addition of state dependent risk-premia. Optimal pricing strategies differ depending on whether firms play Nash or Stackelberg. Furthermore, we find that despite its leadership position, the value function of the leader is lower than the value function of the follower. Finally, we discover that long term profits of competing firms can increase with intensities of the jump processes governing product harm crises. This surprising result happens because anticipating a product harm crisis softens price competition.

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1. Introduction

Product-harm crises hurt all industries. They differ from simple product recalls in that they are discrete and well-publicized occurrences that suddenly force managers to recall massive volume of products (Dawar & Pillutla, 2000; Dawar & Whelan, 2016) and often lead to products' deaths. In the last week of September 2004 for instance, Merck abruptly stopped selling Vioxx. The sudden withdrawal of the drug transformed its initial commercial success into an immediate business nightmare, i.e., sales reached zero in days and the company paid billions of dollars to plaintiffs. The aggravating point was that the firm was aware of this risk but still decided to go ahead with the commercialization of the drug. Product-harm crises happen even when managers do their best to prevent them from occurring, e.g., Mattel recalled over nineteen millions toys due to lead paints in 2007; Johnson & Johnson paid \$3 billion following the recall of its hip replacements in 2013; General Motors recalled over six million cars due to faulty ignition switches in 2014. More recently, Samsung faced a similar crisis. In August 2016, the electronics manufacturer released the Galaxy Note 7, issued a recall for the new device in September 2016 due to faulty batteries and stopped its production in October 2016.

Product-harm crises confront firms to the difficult task of implementing profitable marketing strategies while taking into account the future costs of potential massive recalls. Most marketing studies investigating product-harm crises focus on how brands suffer from such crises and how to manage advertising decisions (e.g., Borah & Tellis, 2016; Dawar & Pillutla, 2000; Eilert, Jayachandran, Kalaighnam, & Swartz, 2017; Gao, Xie, Wang, & Wilbur, 2015; Liu & Shankar, 2015; Roehm & Tybout, 2006; Rubel, Naik, & Srinivasan, 2011; Van Heerde, Helsen, & Dekimpe, 2007). However, to the best of our knowledge, there exists no study that can inform how managers should optimally set prices prior to such rare, yet dramatic events.

Such a problem is dynamic in nature. Indeed, the dates at which crises occur are random making planning horizons stochastic, i.e., product-harm crises can happen very soon after the launch of the product as in the case of Samsung's Galaxy Note, or very late, as in the Vioxx case. As a result managers cannot anticipate ex ante neither how long they will be able to sell the product nor the number of units that will be recalled. The complexity of finding the optimal pricing strategy in such contexts is further enhanced by competition as a tension exists between charging a lower price in response to competition and the resulting impact this lower price can have on the firm's exposure to risk. To shed light on this issue, we propose a differential game where two firms sell differentiated products that can be recalled at any random dates. We model the timing uncertainty of recalls as jump processes, similar

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to Basar and Haurie (1984), and obtain a piecewise deterministic differential game, where the state variables are the number of liability claims that each player would have to manage when confronted to a crisis.

We characterize the optimal pricing strategies and value functions in closed-forms and obtain novel managerial insights. First, we find that the value functions are non-linear in the states, which implies state-dependent strategies. Second, we find that it is always optimal for firms to add a state dependent risk-premium to the prices they would have chosen in the absence of any risk. This result contradicts casual intuition in the sense that a risk prone product would be associated with a lower quality and thus a lower price. This premium equals the shadow price of a marginal unit of potential liabilities that is generated by additional sales. Next, we compare the optimal strategies and value functions in three different scenarios, namely, when there is only one firm in the market, when there are two firms in the market playing Nash and when there are two firms in the market playing Stackelberg. Finally, we find that while the monopolist's profit always goes down as the intensity of the jump process governing the crisis increases, the reverse is not necessarily true in the competitive case, i.e., long-term profits can increase as the intensity of the jump process governing the crisis increases. This last result suggests firms do not always have incentives to allocate resources to make products risk free.

Our results add to the literature on product recalls by taking into the dynamic nature of such phenomena in competitive pricing decisions (e.g., Chao, Irvani, & Savaskan, 2009; Chen, Ganesan, & Liu, 2009; Daughety & Reinganum, 1995). We also contribute to the literature on dynamic pricing in stochastic environments, which focused mostly on random events like competitive entries (e.g., Bensoussan & Sethi, 2007; Chen & Jain, 1992; Fruchter & Messinger, 2003; Kamien & Schwartz, 1971; Rubel, 2013) or recessions (e.g., Caulkins et al., 2011; Huschto et al., 2011), but not on product harm crises. The remainder of the paper evolves as follows. In Section 2, we introduce the model, present the results in Section 3 and conclude in Section 4.

2. Model

We consider a dynamic market where two firms, i.e., $i = \{1, 2\}$, sell differentiated products and compete in price. Both firms face the risk of a product harm crisis, which makes the selling horizon of the two products stochastic. When the focal firm, say i , recalls its product, product i is removed from the market and the competing product remains in the market until it is recalled, e.g., Merck recalled Vioxx in 2004, while Pfizer kept selling Bextra until it was withdrawn in 2005. Consequently, there are two regimes of random durations, i.e., the competitive regime (C) and the monopolistic regime (M), indexed by $R = C$ and $R = M$, respectively. The competitive regime lasts until the first brand is recalled, at which point the system switches to the monopolistic regime until the second brand is recalled. We limit our analysis to monopoly and duopoly markets.

Let time be the continuous variable t and T_1 and T_2 be the dates at which product 1 is recalled and product 2 is recalled, respectively. T_1 and T_2 are random variables that are delineated by the stochastic processes $\{\Gamma_i(t); t \geq 0\}$ for $i = \{1, 2\}$,

$$\lim_{dt \rightarrow 0} \frac{P[\Gamma_i(t+dt) = 1 | \Gamma_i(t) = 0]}{dt} = \chi_i, \quad \lim_{dt \rightarrow 0} \frac{P[\Gamma_i(t+dt) = 0 | \Gamma_i(t) = 1]}{dt} = 0, \quad (1)$$

where χ_i is the intensity of the jump process governing T_i (see e.g., Haurie & Moresino, 2006). Such a specification suggests that while the intensity of the jump process governing a recall is fixed over time, the recall's probability is not. Furthermore, (1) means that

managers do not control the intensities of product harm crises after the product's launch. This assumption comports with empirical research on product harm crises that shows that the root-causes of such dramatic events are determined prior to products' launches, even before brand and product managers make pricing decisions (see, e.g., Thirumalai & Sinha, 2011). In the case of Samsung for instance, the reason of the recall was the defective design of batteries by Samsung's supplier. In 2015, several car manufacturers were also forced to recall vehicles to due Takata's defective airbags. Finally, we note that (1) implies that the intensity of the recall of firm i does not change after the recall of its competitor's product. We make this assumption to keep the model parsimonious without any loss of generality.

We derive products' demands based on Hotelling's model of product differentiation (Hotelling, 1929), which has been widely used in economics and marketing studies. In this model, consumers are uniformly distributed on the unit line and the two brands are located at each ends of the line with Brand 1 on the left and Brand 2 on the right in our case. A consumer located at $y \in [0, 1]$ buys Brand 1 if the utility (U) s/he derives from consuming Brand 1's exceeds the utility derived from consuming Brand's 2, i.e., $1 - p_1 - \alpha y > 1 - p_2 - \alpha(1 - y)$, where α is the misfit cost incurred by consumers. Demands for each brands are identified by the location of the indifferent consumer, that is the consumer located at y such that $1 - p_1 - \alpha y = 1 - p_2 - \alpha(1 - y)$, which yields that $D_i^C(p_i, p_j) = \frac{p_j - p_i + \alpha}{2\alpha}$, with $j = 3 - i$. Similarly, when only one firm is in the market, the product's demand is driven consumers' utilities. Specifically, if Brand 1 is the monopolist after Brand 2's recall, the location of the indifferent consumer is given by $1 - p_1 - \alpha y = 0$, which gives $D_1^M(p_1) = \frac{1 - p_1}{\alpha}$; alternatively, if Brand 2 is the monopolist after Brand 1's recall, the location of the indifferent consumers is given by $1 - p_2 - \alpha(1 - y) = 0$, which gives $D_2^M(p_2) = \frac{1 - p_2}{\alpha}$. Thus, $D_M(p_M) = \frac{1 - p_M}{\alpha}$, where M stands for monopoly and equal 1 (2) if Brand 2 (1) is recalled first.

Everytime a unit of Brand i is sold it increases the number of liability claims that firm i will be exposed to in case of a recall, i.e., $x_i(T_i)$. The dynamics of the states $x_i(t)$ in the competitive and monopolistic regimes are

$$\frac{dx_i}{dt} = D_i^C(p_i, p_j) - \delta x_i(t), \quad \text{with } x_i(0) = 0 \quad (2)$$

and

$$\frac{dx_i}{dt} = D_i^M(p_i) - \delta x_i(t), \quad \text{with } x_i(T_j^+) = x_i(T_j^-), \quad (3)$$

respectively, where $j = 3 - i$, T_j^- refers to the time right before the crisis, and T_j^+ refers to the time right after the crisis. Note that when a crisis happens, firm i recalls $x_i(T_i)$ units, which is not necessarily equal to cumulative sales to date. Specifically, $x_i(T_i)$ can be lower than total sales for various reasons, e.g., buyers might not be aware of the recall; they might not want to take the time to bring the product back to where they bought it; they might already have discarded it before the recall; they might have lost proof of purchase/ownership. We capture this attrition through the parameter $\delta > 0$.

Eq. (2) reveals that the strategic interactions between players are taking place not only through demands, but also through the stocks of "liabilities" accumulated by players, which ultimately impact costs of recalls. Specifically, firms' pricing strategies impact not only instantaneous profits, i.e., $\Pi_i^C(t) = D_i^C(p_i(t), p_j(t))p_i(t)$ but also the stock of liability claims that firms will manage when a recall happens. Firm i 's stock of potential liability claims, i.e., $x_i(t)$, generates a cost to firm i in case of a recall, $k_i(x_i(T_i))$. For simplicity and without any consequence on our results, we do not assume any other additional salvage value or residual profits after the recall of i 's product. Furthermore we set $k_i(x_i(T_i)) = \kappa x_i(T_i)^2$, with

$\kappa > 0$. Consequently, the optimal pricing strategy of firm i is determined such that

$$W_{Ci}(t, x_i, x_j) = \underset{p_i(t)}{\text{Max}} \mathbb{E}_{\Gamma_i, \Gamma_j} \int_0^{\min\{T_i, T_j\}} e^{-rt} \Pi_i^C(t) dt - \kappa x_i(T_i)^2 e^{-rT_i} \mathbf{1}_{[T_i < T_j]} + W_{Mi}(T_j, x_i) e^{-rT_j} \mathbf{1}_{[T_i > T_j]}, \quad (4)$$

with $j = 3 - i$, where the expectation operator is taken with respect to the stochastic processes Γ_i and Γ_j , which governed the random dates of recall. The value function of player i in the competitive regime, i.e., $W_{Ci}(t, x_i, x_j)$, is composed of three terms. The first term is the sum of discounted instantaneous profits that are obtained by firm i for the duration of the competitive regime, i.e., $\int_0^{\min\{T_i, T_j\}} e^{-rt} \Pi_i(t) dt$, with r as the discount factor. The second term, $\kappa x_i(T_i)^2$, captures firm i 's cost of recall if its product were to be the first to be recalled, i.e., $T_i < T_j$ such that $\mathbf{1}_{[T_i < T_j]} = 1$ and $\mathbf{1}_{[T_i > T_j]} = 0$. Finally, the last term, $W_{Mi}(T_j, x_i)$, is the value function obtained by player i if it is its competitor's product that is recalled first, i.e., $T_i > T_j$ such that $\mathbf{1}_{[T_i < T_j]} = 0$ and $\mathbf{1}_{[T_i > T_j]} = 1$ and that the system switches to the monopolistic regime (in favor of player i). This last term equals

$$W_{Mi}(T_j, x_i) = \underset{p_i(t)}{\text{Max}} \mathbb{E}_{\Gamma_i} \left[\int_0^{T_i - T_j} e^{-rt} \Pi_i^M(t) dt - \kappa x_i(T_i)^2 e^{-r(T_i - T_j)} \right], \quad (5)$$

where $\Pi_i^M(t) = D_i^M(p_i(t)) p_i(t)$. Note that in (5), the only random variable remaining is T_i since the recall of the competing product occurred, thus the expectation operator is taken with respect to Γ_i only. Consequently, (1)–(5) define a piecewise deterministic differential game, where the control variables are products' prices and the two state variables the potential liability claims generated by sales. We note that by integration by parts, (4) can be rewritten as an infinite horizon deterministic differential game (see e.g., Boukas, Haurie, & Michel, 1990; Haurie & Moresino, 2006; Rubel, 2013). Over the interval $(t, t + dt)$, the probability that firm i will be forced to recall its product equals χ_i implying that the distribution of T_i equals $f_i(t) = 1 - e^{-\chi_i t}$. Hence, (4) becomes $W_{Ci} = \underset{p_i(t)}{\text{Max}} \{I_1 + I_2\}$,

where $I_1 = \int_0^\infty \chi_i e^{-\chi_i s} e^{-\chi_j s} e^{-rs} \left\{ \int_0^t \Pi_i^C(\eta) d\eta - \kappa x_i(t) \right\} ds$ and $I_2 = \int_0^\infty \chi_j e^{-\chi_i s} e^{-\chi_j s} e^{-rs} \left\{ \int_0^t \Pi_i^C(\eta) d\eta + W_{Mi}(t, x_i) \right\} ds$, which after algebraic manipulations yields

$$W_{Ci}(x_i, x_j) = \underset{p_i(t)}{\text{Max}} \int_0^\infty e^{-(\chi_i + \chi_j + r)t} \left\{ \Pi_i^C(t) - \chi_i \kappa x_i^2 + \chi_j W_{Mi}(t, x_i) \right\} dt. \quad (6)$$

Thus, the Hamilton–Jacobi–Bellman (HJB) equation of player i in the competitive regime is

$$rW_{Ci}(x_i, x_j) = \underset{p_i}{\text{Max}} \left\{ \Pi_i^C(p_i, p_j) - \chi_i (\kappa x_i^2 - W_{Ci}(x_i, x_j)) + \chi_j (W_{Mi}(x_i) - W_{Ci}(x_i, x_j)) + \frac{\partial W_{Ci}}{\partial x_i} \frac{dx_i}{dt} + \frac{\partial W_{Ci}}{\partial x_{3-i}} \frac{dx_{3-i}}{dt} \right\}, \quad (7)$$

which comports with Dockner, Jørgensen, Long, and Sorger (2000). Similarly, (5) can be transformed into

$$W_{Mi}(x_i) = \underset{p_i(t)}{\text{Max}} \int_0^\infty e^{-(\chi_i + r)t} \left\{ \Pi_i^C(t) - \chi_i \kappa x_i^2 \right\} dt, \quad (8)$$

yielding the HJB equation

$$rW_{Mi} = \underset{p_i}{\text{Max}} \left\{ \Pi_i^M(p_i) - \chi_i (\kappa x_i^2 + W_{Mi}) + \frac{\partial W_{Mi}}{\partial x_i} \frac{dx_i}{dt} \right\}, \quad (9)$$

which is also consistent with Dockner et al. (2000). We see that the jump intensities enter the firms' objective functions at two different places. On the one hand, jump intensities allow firms to take into account expected recall costs in their “instantaneous profits”. On the other hand, the jump intensities increase managers' “impatience,” i.e., $e^{-(\chi_i + \chi_j + r)t}$ in (6) and $e^{-(\chi_i + r)t}$ in (8), respectively. Next, we characterize the optimal pricing strategies when firms play Nash and then characterize the pricing strategies when firms play Stackelberg.

3. Results

To do so, we proceed backward and first characterize the value function of player i in the monopoly regime, that is after the product of player $j = 3 - i$ has been recalled.

3.1. Monopoly regime

We differentiate (9) with respect to the control variable p_i and equating the resulting expression to zero, we find that the optimal pricing strategy in the monopolistic regime equals $p_i(x_i)^M = \frac{1}{2} - \frac{1}{2} \frac{\partial W_{Mi}}{\partial x_i}$, that is

$$p_i(x_i)^M = p_{RF}^M - \frac{1}{2} \frac{\partial W_{Mi}}{\partial x_i}. \quad (10)$$

Eq. (10) reveals that the optimal pricing strategy for a risk-prone product is composed of two terms, namely, the price that should be charged for a risk-free product in a monopoly market, i.e., p_{RF}^M , and the shadow price of a marginal unit of liability, i.e., $\frac{\partial W_{Mi}}{\partial x_i}$, which allows us to obtain the following proposition.

Proposition 1. *The optimal pricing strategy and value function in the monopoly regime are*

$$p_M^*(x_i) = \frac{1 - (M_2 x_i + M_1)}{2} \quad (11)$$

and

$$W_M(x_i) = \frac{M_2}{2} x_i^2 + M_1 x_i + M_0, \quad (12)$$

respectively, where

$$M_2 = \alpha \left(2\delta + \rho + \chi - \sqrt{\frac{4\kappa\chi + \alpha(2\delta + \rho + \chi)^2}{\alpha}} \right) < 0 \quad (13)$$

$$M_1 = -\frac{M_2}{M_2 - 2\alpha(\delta + \rho + \chi)} < 0 \quad (14)$$

$$M_0 = \frac{(1 + M_1)^2}{4\alpha(\rho + \chi)} > 0 \quad (15)$$

Proof. See appendix. $\square$

Proposition 1 reveals that the firm adds a risk premium to the price it would have charged for a risk-free product, which yields that $p_i^M > p_{RF}^M$. The risk premium is equal to the sensitivity of the value function with respect to $x_i(t)$ and it achieves two objectives. First, it allows the firm to provision financial resources to manage the cost of future potential contingencies generated by the recall. This insight provides a theoretical foundation to the Financial Accounting Standards number 5 from the Financial Accounting Standards Board that requires companies to include in the financial information provided to investors, information about future contingencies that might affect firms in the future, and in particular threats of product (see e.g., Miller & Narayanan, 2002). The second role played by the risk premium is to increase price and consequently to limit the number of products put on the market. Thus,

the risk premium empowers the firm to optimally balance profitability and risk. Furthermore, Proposition 1 reveals that the risk premium is state dependent, thus making the optimal price varying with x_i such that $\frac{\partial p_M^*}{\partial x_i} > 0$. Additional comparative statics reveal that $\frac{\partial p_M^*}{\partial \chi} > 0$, $\frac{\partial p_M^*}{\partial \delta} < 0$ and $\frac{\partial p_M^*}{\partial \kappa} > 0$.

Finally, we find that $\frac{\partial W_M^N(x_i)}{\partial \chi} < 0$, which means that the monopolist's long-term profit (or value function) unambiguously decreases as the intensity of the jump process governing the recall increases. While this result seems intuitive, we will show that this is not always the case in the competitive regime. To that end, we next characterize the optimal pricing strategies and value functions in a duopoly market when firms play either Nash or Stackelberg.

3.2. Competitive regime when firms play Nash

Proposition 2. The optimal Nash pricing strategies and value functions in the competitive regime are

$$p_i^*(x_i, x_j) = \alpha + \frac{2\left(-\frac{\partial W_{Ci}^N(x_i, x_j)}{\partial x_i} + \frac{\partial W_{Cj}^N(x_i, x_j)}{\partial x_j}\right) + \frac{\partial W_{Ci}^N(x_i, x_j)}{\partial x_i} - \frac{\partial W_{Cj}^N(x_i, x_j)}{\partial x_j}}{3} \quad (16)$$

and

$$W_{Ci}^N(x_i, x_j) = \frac{A_{1i}}{2} x_i^2 + A_{2i} x_i + A_{3i} x_i x_j + A_{4i} x_j + \frac{A_{5i}}{2} x_j^2 + A_{6i}, \quad (17)$$

respectively, with $i = \{1, 2\}$ and $j = 3 - i$, where the value functions' coefficients are obtained by solving a system of twelve non linear equations, provided in the appendix.

Proof. See appendix. $\square$

The optimal pricing strategies remain state dependent in the competitive regime, as it should be given the quadratic nature of the value functions. The structure of the risk premium added by player i , however, is qualitatively different than in the monopoly case. Specifically, the risk premium is obtained by the addition of two terms of symmetric nature, i.e., $2\left(-\frac{\partial W_{Ci}^N(x_i, x_j)}{\partial x_i} + \frac{\partial W_{Cj}^N(x_i, x_j)}{\partial x_j}\right)$ and $\frac{\partial W_{Ci}^N(x_i, x_j)}{\partial x_i} - \frac{\partial W_{Cj}^N(x_i, x_j)}{\partial x_j}$. The first term incorporates not only how the price of product i impacts x_i , but also how it impacts x_j , which means that when choosing strategic prices when envisioning product harm crises, managers must take into account how their prices will affect the stock of liability claims of competing brands. Said differently, ignoring this effect will lead to profit losses. The second term is similar in that it directly comes from the strategic interactions between the two prices. Thus, contrary to risk premia in the monopoly regime, risk premia in the competitive regime must account how prices affect all competing brands in the market.

To gain further insights about the equilibrium and its sensitivity to the key parameters, we provide in Corollary 1 the optimal pricing strategies and value functions in closed form when firms are symmetric, that is by assuming that players' specific parameters are identical, which yields that $A_{ki} = A_{kj} = A_k$, with $k = 1, 2, \dots, 6$.

Corollary 1. When firms are symmetric, the optimal Nash pricing strategies and value functions in the competitive regime are

$$p_i^*(x_i, x_j) = \alpha - A_2 + A_4 + \frac{(-2A_1 + A_3 + A_5)x_i - (A_1 + A_3 - 2A_5)x_j}{3} \quad (18)$$

and

$$W_{Ci}^N(x_i, x_j) = \frac{A_1}{2} x_i^2 + A_2 x_i + A_3 x_i x_j + A_4 x_j + \frac{A_5}{2} x_j^2 + A_6, \quad (19)$$

respectively, with $i = \{1, 2\}$ and $j = 3 - i$, where

$$A_1 = 3\left(A_3 - \sqrt{-A_3\alpha(2\delta + \rho + 2\chi)}\right) \quad (20)$$

$$A_2 = 3\frac{A_2 - M_1\chi - \sqrt{-A_3\alpha(2\delta + \rho + 2\chi)}}{\delta + \rho + 2\chi} \quad (21)$$

$$A_3 = -\frac{8(M_2 + 2\kappa)\chi(2\delta + \rho + 2\chi) + 9\alpha(2\delta + \rho + 2\chi)^3}{32(2\delta + \rho + 2\chi)^2} - \frac{3\sqrt{\alpha}(2\delta + \rho + 2\chi)^4(16(M_2 + 2\kappa)\chi + 9\alpha(2\delta + \rho + 2\chi)^2)}{32(2\delta + \rho + 2\chi)^2} \quad (22)$$

$$A_4 = \frac{-3A_3 + \sqrt{-A_3\alpha(2\delta + \rho + 2\chi)}}{\delta + \rho + 2\chi} \quad (23)$$

$$A_5 = \frac{3\sqrt{\alpha}(2\delta + \rho + 2\chi)^4(16(M_2 + 2\kappa)\chi + 9\alpha(2\delta + \rho + 2\chi)^2)}{32(2\delta + \rho + 2\chi)^2} + \frac{8(M_2 + 2\kappa)\chi(2\delta + \rho + 2\chi) + 9\alpha(2\delta + \rho + 2\chi)^3}{32(2\delta + \rho + 2\chi)^2} \quad (24)$$

$$A_6 = \frac{2A_4 + \alpha - 2M_0\chi}{2(\rho + 2\chi)} \quad (25)$$

where M_2 , M_1 and M_0 are provided in Proposition 1.

Proof. Follows directly from Proposition 2. Note that we kept the roots for the coefficients of the value function that guarantee a unique stable steady state. $\square$

Note that contrary to the results in Proposition 2, we are now able to solve the coefficients of the value functions in closed forms. Next, we conduct numerical simulations to learn how pricing strategies vary with key parameters. These analyses were performed using Mathematica 10.1.0 and presented below.

Panel A reveals that firms should implement higher pricing for any level of sales as the intensity of a product harm crisis increases. Similarly, Panel B reveals that firms should implement higher pricing for any level of sales as the cost parameter κ increases. These two results comport with our findings in the monopoly case. They are intuitive as they suggest that firms should increase prices as the expected cost of a product harm crisis increases. Similarly, Panel C of Fig. 1 suggests that as α increases, firms can charge higher prices, which is also intuitive as α measures to some extent the degree of differentiation between the two brands in the marketplace. Finally, results from Panel D also comport with economic intuition, since δ measures the decay rate of liabilities claim the firms can face in case of recall. Next, we investigate how the value functions vary with important model's parameters.

We first provide Fig. 2 which depicts how the value function varies in the state variables

Second, since the value function decreases in the the state variable, we conduct further numerical analyses on the value functions at $t = 0$, that is at the beginning of the game. This choice is driven by the desire to understand how at the beginning of the game, firms' long term profits (i.e., value functions) vary with model's parameters and in particular χ . We report the results in Fig. 3.

As what could be expected, the value functions are decreasing in the cost parameter κ (see Panel A), and increasing in α (see Panel B), which is intuitive. However, Fig. 3 also reveals two counter intuitive insights. First, it reveals that the value functions increase then decrease as χ increases. Moreover, it reveals that the

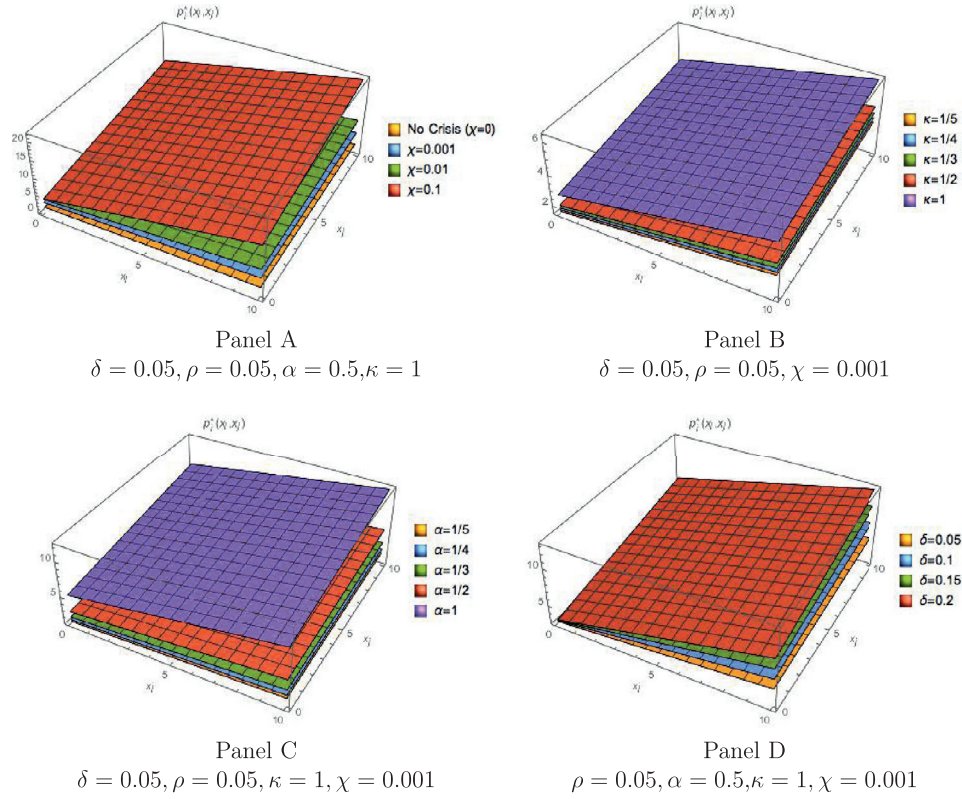


Fig. 1. Pricing strategies.

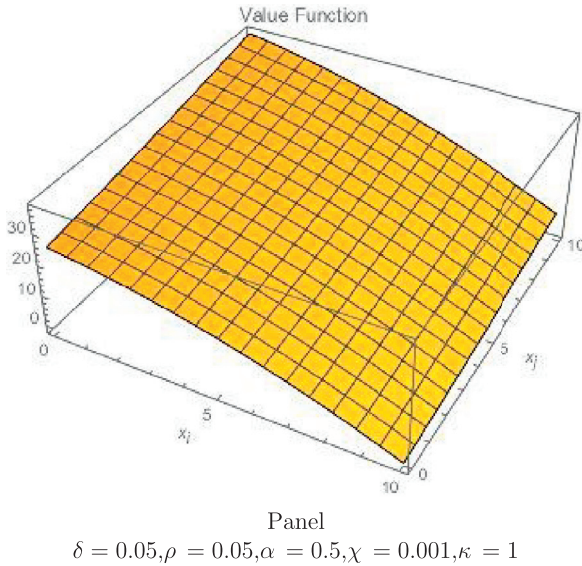


Fig. 2. Value function.

value function of a risk-prone product can exceed the long-term profit that would be obtained by a risk free product, that is when $\chi = 0$. Said differently, firms can profit from product-harm crises.

The intuition for these two novel insights follows from the fact that prices are higher for risk-prone products than they are for risk free products. Thus, what Figs. 1 and 3 suggest is that the threat of facing a product recall softens price competition, which can lead to higher profit in the competitive regime. To confirm this intuition, we perform similar analyses with identical parameter values in a market with only one firm. Specifically, we conduct our

numerical simulations on the value function provided in Proposition 1, assuming that the state variable is zero and obtain Fig. 4, which corroborates the analytical insight obtained earlier that the monopolist's value function always decreases in χ .

Thus profits of firms selling risk prone products can exceed profits of firms selling risk free products. This counter intuitive result comes from the fact that the mere likelihood of a crisis softens price competition. This finding has also implications on firms' incentives to reduce (or not) the likelihood of a crisis. Specifically, Fig. 3 suggests that it can be optimal for managers to sell differentiated products that are not risk free. Next, we revisit these results assuming that firms play Stackelberg.

3.3. Competitive regime when firms play Stackelberg

We characterize the Feedback Stackelberg equilibrium (Jørgensen & Zaccour, 2004) assuming that player 2 is the leader and player 1 the follower and present the following proposition.

Proposition 3. The optimal Stackelberg pricing strategies for the follower and the leader are

$$p_1^*(x_1, x_2) = \frac{\alpha + p_2^* - \frac{\partial W_{C1}^S}{\partial x_1} + \frac{\partial W_{C1}^S}{\partial x_2}}{2} \quad (26)$$

and

$$p_2^*(x_1, x_2) = \frac{3\alpha - \frac{\partial W_{C1}^S}{\partial x_1} + \frac{\partial W_{C1}^S}{\partial x_2} + \frac{\partial W_{C2}^S}{\partial x_1} - \frac{\partial W_{C2}^S}{\partial x_2}}{2} \quad (27)$$

respectively. The value function of the follower and the leader are

$$W_{C1}^S(x_1, x_2) = \frac{a_1}{2}x_1^2 + a_2x_1 + a_3x_1x_2 + a_4x_2 + \frac{a_5}{2}x_2^2 + a_6 \quad (28)$$

and

$$W_{C2}^S(x_1, x_2) = \frac{b_1}{2}x_1^2 + b_2x_1 + b_3x_1x_2 + b_4x_2 + \frac{b_5}{2}x_2^2 + b_6 \quad (29)$$

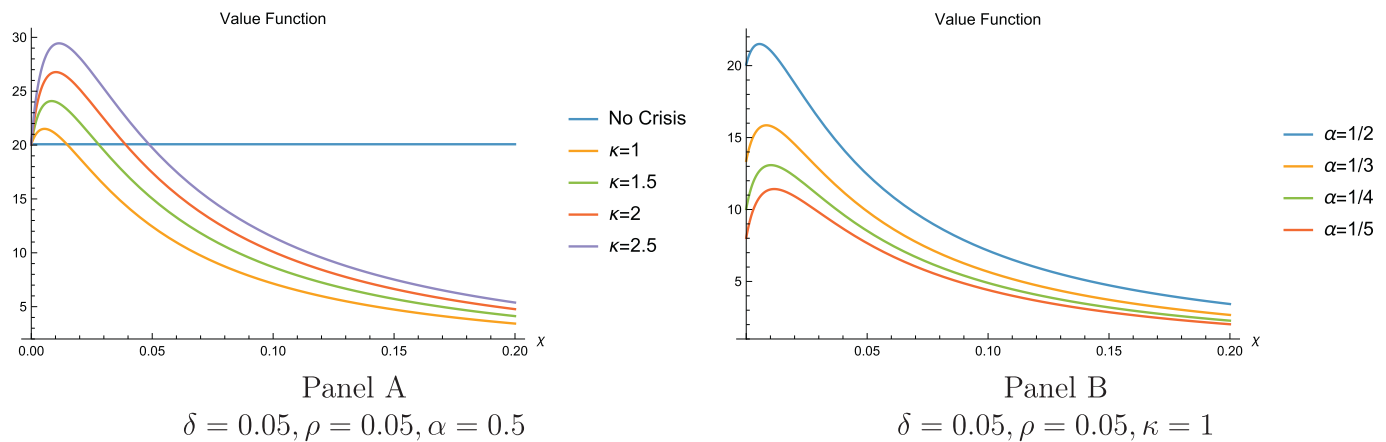


Fig. 3. Value function in the competitive case.

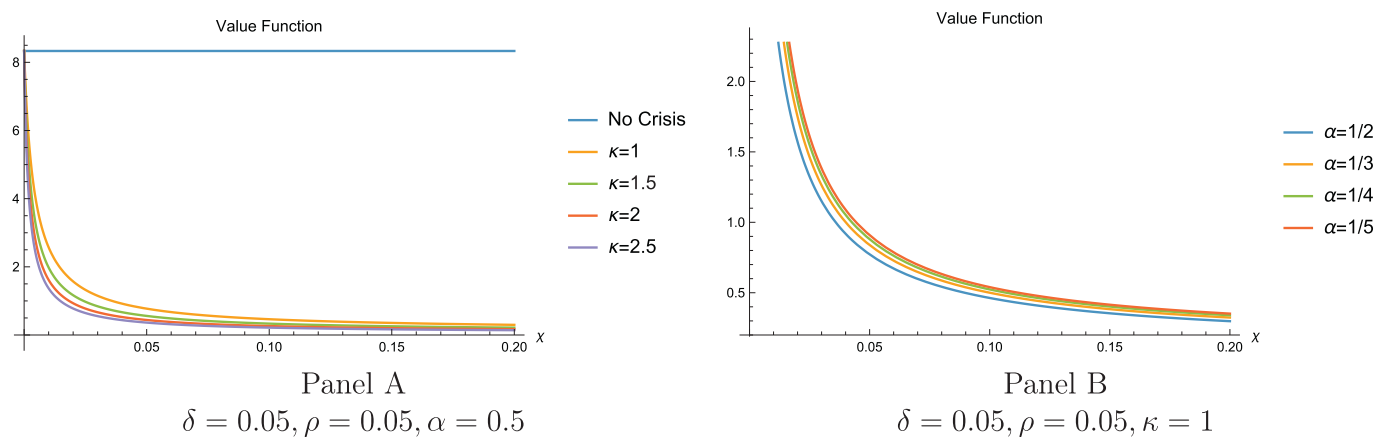


Fig. 4. Value function in the monopoly case.

respectively, where the value functions' coefficients are obtained by solving a system of twelve non linear equations, provided in the appendix.

Proof. See appendix. $\square$

This new Proposition furnishes the characterization of the Feedback Stackelberg equilibrium for two firms competing in price and who face the threat of product harm crises. First, we note that the Nash and Stackelberg equilibria do not coincide. Furthermore, similar to the previous section, we conduct numerical analyses to gain further knowledge on the optimal pricing strategies for both the Leader and the Follower, as well as on their respective long term profits. To this end, we solve the system of twelve non linear equations that characterize the coefficients of the value function numerically with the procedure NSolve in Mathematica. We solve this system such that to retain solutions that guarantee asymptotically stable equilibria. Fig. 5 displays the optimal pricing strategies, while Fig. 6 provides the value functions.

Results in Fig. 6 comport with the findings obtained in for the Nash equilibrium, that is, i.e., the intensity of the crisis increases (decreases) the firms' value functions for low (high) levels of χ and it is possible for both the Stackelberg leader and the Stackelberg follower to obtain higher long term profits than in a risk free market. This generalization provides some robustness check for the results obtained when firms play Nash. Figs. 5 and 6 also reveal two new insights. They show that the leader's value function is lower than the follower's, which suggests that when firms face a risk of product-harm crisis, being the follower is beneficial. Why?

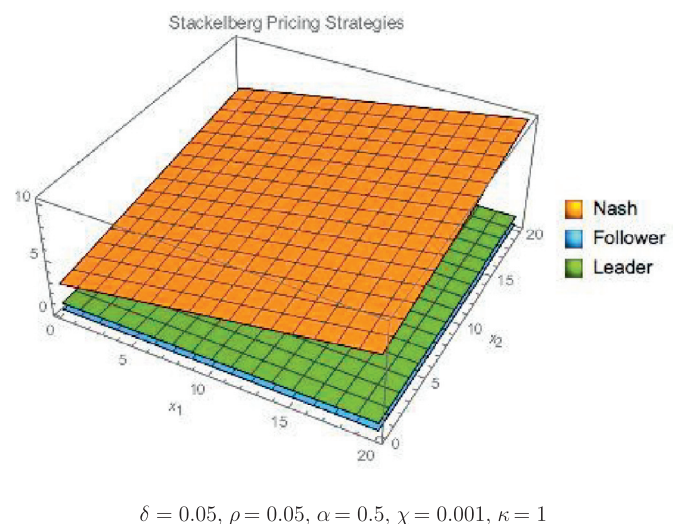


Fig. 5. Stackelberg pricing strategies.

Because the leader uses its leadership position to charge a higher price than the follower. By doing that, the leader reduces the firm's exposure to risk more than the follower. On the contrary, the follower charges a lower price, which translates into more sales and hence profits.

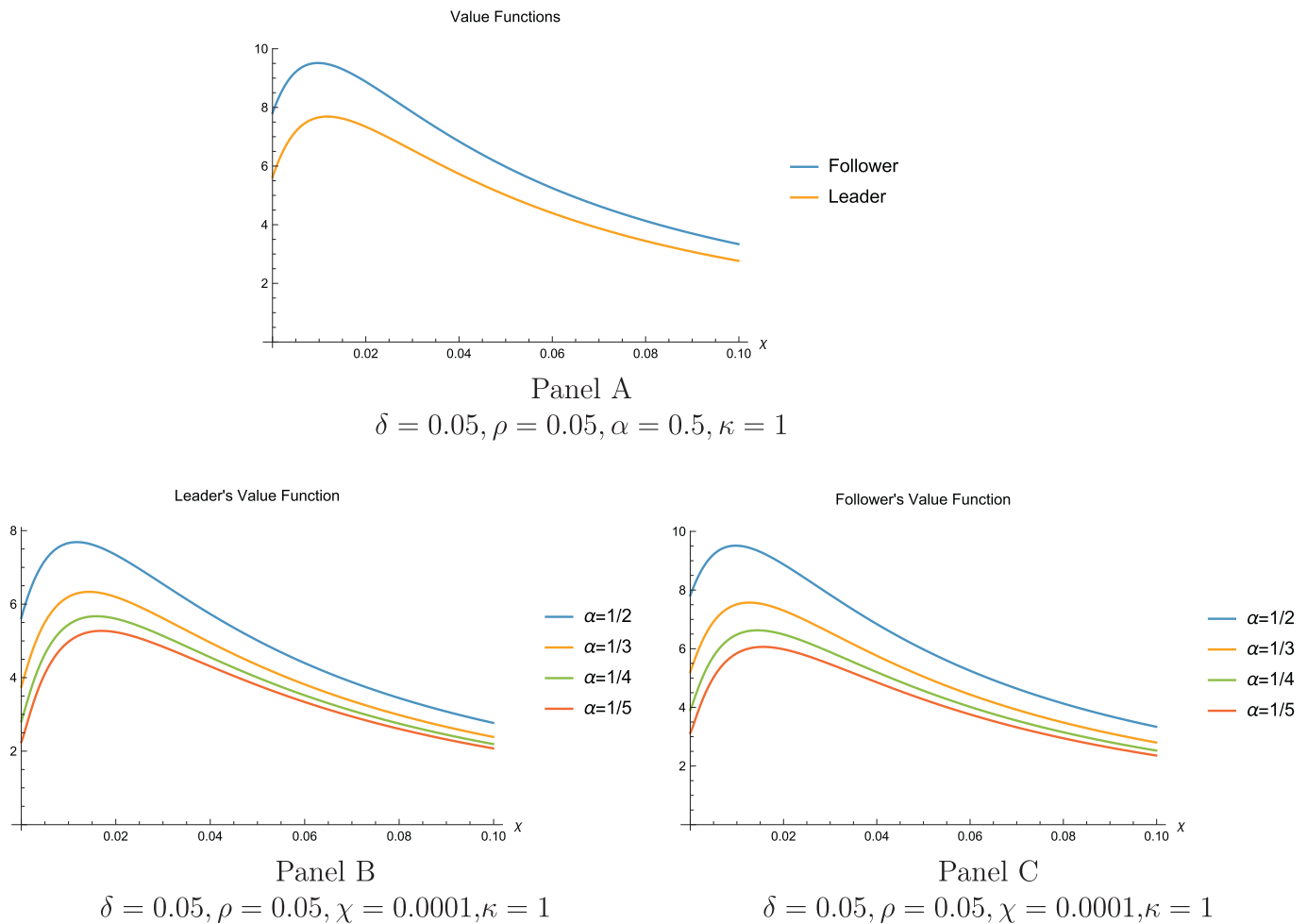


Fig. 6. Value functions in the Stackelberg game.

4. Concluding remarks

The proposed model sheds lights on firms' pricing decisions when envisioning product harm crises. In our model, firms compete in price and do not know when a product harm crisis will come and who is going to be affected. If the focal firm is affected by the crisis, it has to recall products and pay liability claims, which over time increase with sales but decrease with attrition. Since the date of the recall is uncertain, the volume of claims that the focal firm will have to handle is also uncertain. If the focal firm's product is not recalled, but it is instead its competitor's product that is affected by the product-harm crisis, then the focal firm becomes a monopolist as the competitor's product is withdrawn from the market. We model this game as a piecewise deterministic differential game to capture the timing uncertainty around when crises happen and also the change in market competition due to recalls. Furthermore, we explore how pricing strategies and long-term profits, as measured by players' value functions, vary with the mode of play of the game and obtain several results.

First, we find that pricing risky products requires to add risk premia to prices that would be chosen in the absence of the risk of product harm crises. Next, we find that Nash and Stackelberg equilibria do not coincide. In particular, we find that firms choose lower prices when they play Nash, compared to when they play Stackelberg. Moreover, we find that the Stackelberg's leader exploits its leadership position by charging a higher price than the

follower. However, we find that the leadership position in this game does not allow the leader to capture more profit. On the contrary, we discover that the follower's value function exceeds the leader's. This result could be explained by the dual role played by jump intensities in the firms' objective functions. Specifically, jump intensity enters firms' objective functions through the expected costs of recall and through managers' enhanced impatience.

The most surprising results that are obtained thank to the proposed model is that firms can profit from the risk of a product harm crisis, but only when the market is competitive. We find that the intensity of a crisis, χ in our model, can not only increase the value functions of all players, but can also allow firms to capture more profits than when risk free products compete. To explain this counter intuitive result, we suggest that the mere anticipation of a crisis softens price competition and thus generates more profits.

The proposed model can be extended in several directions. First, we assumed that the only variable used by firms to generate demand and control exposure to possible liabilities is price. In practice, demand is generated by other actions, e.g., advertising, and it would be interesting to understand how firms should dynamically manage their marketing mix when envisioning product harm crises. Second, we assumed that products were sold directly to end consumers. However, in many industries, products are sold through a network of intermediaries, e.g., car dealers or retailers. Thus, investigating the role of contractual agreements in marketing channels when there is a threat of a product-harm crisis would be of great managerial and methodological interests.

Appendices

Proof Proposition 1

After substituting (10) in (9), we conjecture that the value function is linear in the state, i.e., $W_{Mi}(x_i) = \frac{M_2}{2}x_i^2 + M_1x_i + M_0$. We obtain the coefficients of the value function by applying the method of undetermined coefficients. There are two possible roots for M_2 , i.e., $\alpha \left(2\delta + \rho + \chi + \sqrt{\frac{4\kappa\chi + \alpha(2\delta + \rho + \chi)^2}{\alpha}} \right)$ and $\alpha \left(2\delta + \rho + \chi - \sqrt{\frac{4\kappa\chi + \alpha(2\delta + \rho + \chi)^2}{\alpha}} \right)$. We retain $M_2 = \alpha \left(2\delta + \rho + \chi - \sqrt{\frac{4\kappa\chi + \alpha(2\delta + \rho + \chi)^2}{\alpha}} \right)$, which ensures stability of the equilibrium.

Proof Proposition 2

Differentiating (7) with respect to p_i , we solve for $\{p_i, p_j\}$ and obtain the first order conditions $p_i^*(x_i, x_j) = 2 \left(-\frac{\partial W_{Ci}^N(x_i, x_j)}{\partial x_i} + \frac{\partial W_{Ci}^N(x_i, x_j)}{\partial x_j} \right) + \frac{\partial W_{Cj}^N(x_i, x_j)}{\partial x_i} - \frac{\partial W_{Cj}^N(x_i, x_j)}{\partial x_j}$. Next, we replace the FOCs in (7), conjecture that the value functions are $W_{Ci}^N(x_i, x_j) = \frac{A_{1i}}{2}x_i^2 + A_{2i}x_i + A_{3i}x_i x_j + A_{4i}x_j + \frac{A_{5i}}{2}x_j^2 + A_{6i}$, and finally apply the method of undetermined coefficients to obtain a system of 12 nonlinear equations that characterize the Nash equilibrium. The system of equations is

$$A_{1i} = \frac{A_{1i}^2 + (A_{3i} + A_{3j} - A_{5j})^2 - 2A_{1i}(A_{3i} - A_{5j} + A_{3j} + 9\alpha\delta) - 9\alpha(M_2 + 2\kappa)\chi}{9\alpha(\rho + 2\chi)} \quad (30)$$

$$\begin{aligned} (\rho + 2\chi)A_{2i} &= (A_{3i} + A_{3j} - A_{5j})(A_{4i} - A_{4j} + A_{2j}) \\ &\quad + 3\alpha(2A_{3i} + A_{5j} - A_{3j}) \\ &\quad \times A_{1i}(A_{2i} - A_{4i} + A_{4j} - A_{2j} + 3\alpha) \\ &\quad - A_{2i}(A_{3i} - A_{5j} + A_{3j} + 9\alpha\delta) \end{aligned} \quad (31)$$

$$A_{3i} = \frac{(A_{1i} - A_{3i} - A_{3j} + A_{5j})(A_{3i} - A_{5i} + A_{3j} - A_{1j}) - 18\alpha\delta A_{3i}}{9\alpha(\rho + 2\chi)} \quad (32)$$

$$\begin{aligned} (\rho + 2\chi)A_{4i} &= (A_{4i} - A_{4j} + A_{2j})(A_{5i} - A_{3j}) + A_{2i}(A_{3i} - A_{5i} + A_{3j} - A_{1j}) \\ &\quad - A_{3i}(A_{4i} - A_{4j} + A_{2j} - 3\alpha) \\ &\quad \times A_{1j}(A_{4i} - A_{4j} + A_{2j} - 3\alpha) \\ &\quad + 3(2A_{5i} + A_{3j})\alpha - 9\alpha\delta A_{4i} \end{aligned} \quad (33)$$

$$(\rho + 2\chi)\frac{A_{5i}}{2} = \frac{(A_{3i} - A_{5i} + A_{3j} - A_{1j})^2}{18\alpha} - 5\delta A_{5i} \quad (34)$$

$$(\rho + 2\chi)A_{6i} = \frac{(A_{2i} - A_{4i} + A_{4j} - A_{2j})^2 + 6(A_{2i} + 2A_{4i} + A_{4j} - A_{2j})\alpha + 9\alpha^2}{18\alpha} - M_0\chi \quad (35)$$

where $i = 1, 2$ and where the expressions for M_2 , M_1 and M_0 are provided in Proposition 1.

Proof Proposition 3

We follow Jørgensen and Zaccour (2004) to characterize the Feedback Stackelberg equilibrium. We assume that player 2 is the leader and player 1 is the follower. We first differentiate (7) with respect to p_1 to obtain the first order condition of the follower. Next, we insert player 1's FOC in player 2's HJB equation and then differentiate it with respect to p_2 . The, we insert these FOCs into player 1 and player 2's HJB equations and conjecture that the value functions are $W_{C1}^S(x_1, x_2) = \frac{a_1}{2}x_1^2 + a_2x_1 + a_3x_1x_2 + a_4x_2 + \frac{a_5}{2}x_2^2 + a_6$ and $W_{C2}^S(x_1, x_2) = \frac{b_1}{2}x_1^2 + b_2x_1 + b_3x_1x_2 + b_4x_2 + \frac{b_5}{2}x_2^2 + b_6$. Finally we apply the method of undetermined coefficients to obtain a system of 12 nonlinear equations that characterize the Feedback Stackelberg equilibrium, i.e.,

$$\begin{aligned} (\rho + 2\chi)\frac{a_1}{2} &= \\ \frac{a_1^2 + (a_3 - b_1 + b_3)^2 - 2a_1(a_3 - b_1 + b_3 + 16\alpha\delta) - 16\alpha(M_2 + 2\kappa)}{32\alpha} \end{aligned} \quad (36)$$

$$\begin{aligned} 16\alpha(\rho + 2\chi)a_2 &= a_4(a_3 - b_1 + b_3 - a_1) + (b_3 - b_1 - a_1 + a_3) \\ &\quad \times (b_4 - a_2 - b_2 - 5\alpha) \\ &\quad - 16\alpha(a_2\delta - a_3) - 16M_1\alpha\chi \end{aligned} \quad (37)$$

$$(\rho + 2\chi)a_3 = \frac{(a_1 - a_3 + b_1 - b_3)(a_3 - a_5 + b_3 - b_5)}{16\alpha} - 2a_3\delta \quad (38)$$

$$\begin{aligned} 16\alpha(\rho + 2\chi)a_4 &= (a_2 + b_2 - b_4 - a_4)(a_3 - a_5 + b_3 - b_5) \\ &\quad + \alpha(5(a_3 + b_3 - b_5) + 11a_5 - 16a_4\delta) \end{aligned} \quad (39)$$

$$(\rho + 2\chi)\frac{a_5}{2} = \frac{(a_3 - a_5 + b_3 - b_5)^2}{32\alpha} - a_5\delta \quad (40)$$

$$\begin{aligned} 32\alpha(\rho + 2\chi)a_6 &= (b_2 - b_4 + 5\alpha)^2 + (a_2 - a_4)^2 \\ &\quad + 2(a_2 - a_4)(b_2 - b_4) \\ &\quad + 2\alpha(5a_2 + 11a_4 - 16\chi M_0) \end{aligned} \quad (41)$$

$$(\rho + 2\chi)\frac{b_1}{2} = \frac{(a_1 - a_3 + b_1 - b_3)^2}{16\alpha} - b_1\delta \quad (42)$$

$$(\rho + 2\chi)b_2 = \frac{(a_2 - a_4 - b_4 + b_2 - 3\alpha)(a_1 - a_3 + b_1 - b_3) - 8\alpha(b_1 + \delta b_2)}{8\alpha} \quad (43)$$

$$(\rho + 2\chi)b_3 = \frac{(a_1 - a_3 + b_1 - b_3)(a_3 - a_5 + b_3 - b_5)}{8\alpha} - 2b_3\delta \quad (44)$$

$$(\rho + 2\chi)b_4 = \frac{(a_2 - a_4 + b_2 - b_4 - 3\alpha)(a_3 - a_5 + b_3 - b_5) - 8\alpha(b_3 + b_4 + \chi M_1)}{8\alpha} \quad (45)$$

$$(\rho + 2\chi)\frac{b_5}{2} = \frac{((a_3 - a_5) - (b_3 - b_5))^2 - 8\alpha(2\delta b_5 + \chi(M_2 + 2\kappa))}{16\alpha} \quad (46)$$

$$(\rho + 2\chi)b_6 = \frac{((a_2 - a_4) - (b_2 - b_4))^2 - 2\alpha(8M_0\chi + 3(a_2 - a_4))}{16\alpha} \quad (47)$$

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